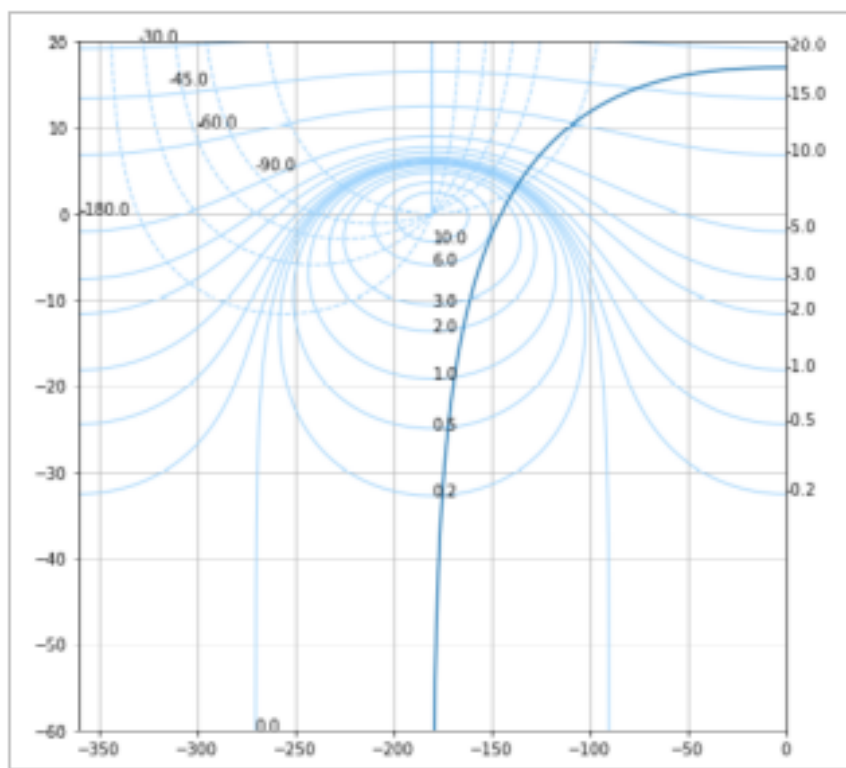
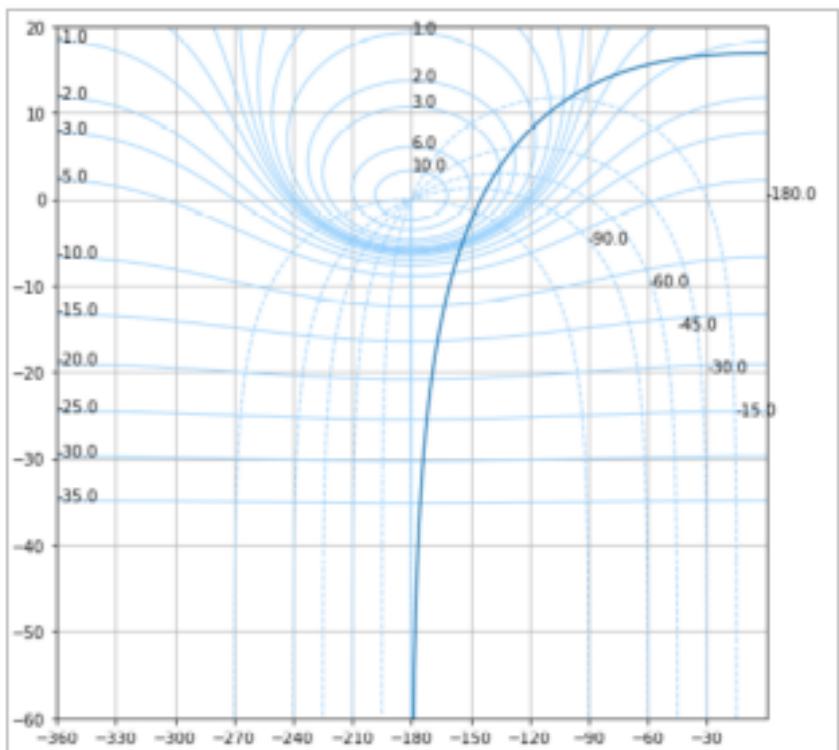


University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problem Set 5.3: Frequency Domain Analysis

Question 1

Consider the following Nichols and Inverse Nichols charts for a second-order system:

- A) Is the system stable? Give rough gain and phase margins, and explain whether you think the system's stability is robust.
- B) Roughly sketch the shape of the magnitude Bode plot for the closed-loop system. Based on your sketch, describe the damping characteristics you expect the closed-loop system's response to have, and whether it has complex poles.
- C) Another engineer warns you that the plant's behaviour is susceptible to temperature-related fluctuations. Do you expect that the closed-loop system will respond consistently despite this problem? What about high-frequency noise?
- D) Do you agree that the system would be more robust if a smaller gain was used? Would your answer change if you were told that it is to be used exclusively for very low frequency operation?



Question 2

A system has a type number greater than zero. With reference to its gain at low frequencies (< 1 rad/s) explain why it might be easier to analyse using Bode or Nichols methods, rather than a Nyquist plot.

Question 3

Use a program of your choice to construct Bode, Nyquist, Nichols and Inverse Nichols plots for the following systems:

- i) $\frac{582.9}{s^3+18s^2+95s+126}$
- ii) $\frac{39.27s^2+392.7s+1335}{s^3+20s^2+141s+360}$
- iii) $\frac{229.6s-229.6}{s^4+11s^3+46s^2+86s+60}$

Based on your plots, write a short assessment of the system considering the following characteristics:

- A) **Stability.** If it is stable, identify its stability margins, and explain whether you perceive them to be an accurate reflection of how stable the system is. If it is not, evaluate whether it can be stabilized by changing the gain.
- B) **Frequency response:** describe whether the system amplifies or attenuates low frequencies, high frequencies and any other frequency ranges you find interesting. Is the system vulnerable to high-frequency noise? If the system is unstable and can be stabilized, you may consider the stabilized version.
- C) Repeat (B), but for the closed-loop response.
- D) **Robustness:** are gain or phase fluctuations more dangerous for the system? Do you expect the system's behaviour to remain consistent if the open-loop model's parameters fluctuate? Again, you may consider a stabilized model if necessary.